

Daily Geometry Drill #1 — Answers & Investigations

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.

Section A Rapid Recognition

1. Find the gradient of the line $6x - 3y + 15 = 0$.

Answer: 2

Method: Rearrange to gradient–intercept form: $3y = 6x + 15 \Rightarrow y = 2x + 5$, so $m = 2$.

Investigate further: Implicit form $ax + by + c = 0$ has gradient $-a/b$; here $-\frac{6}{3} = 2$. **If the line is instead written $3y - 6x - 15 = 0$ (the same line), is the gradient still 2?** It must be — check the $-a/b$ rule still agrees.

2. Find the gradient of the line through $(1, 2)$ and $(5, 14)$.

Answer: 3

Method: $m = \frac{14-2}{5-1} = \frac{12}{4} = 3$.

Investigate further: Check with $(5 - 1, 14 - 2)$ swapped: $\frac{2-14}{1-5} = \frac{-12}{-4} = 3$ ✓ — the sign always cancels. **If the line through $(1, 2)$ and $(5, 14)$ is moved up by 3 (every y increased by 3), does its gradient change?** No — a vertical translation keeps the ratio of rises fixed.

3. A line of gradient 4 is perpendicular to a line of gradient m . Write down m .

Answer: $-\frac{1}{4}$

Method: Perpendicular gradients are negative reciprocals: $m = -\frac{1}{4}$, since $4 \cdot (-\frac{1}{4}) = -1$.

Investigate further: The product test $m_1 m_2 = -1$ is the fastest check. **Is the line $y = 4x$ parallel, perpendicular, or neither, to $y = -\frac{1}{4}x + 12$?** The constant term is a distraction — gradients alone decide.

4. Find the distance between $(2, 1)$ and $(5, 5)$.

Answer: 5

Method: $d = \sqrt{(5-2)^2 + (5-1)^2} = \sqrt{9+16} = \sqrt{25} = 5$.

Investigate further: 3-4-5 right triangle in disguise. **The points $(-1, 0)$ and $(3, -3)$ are the same distance apart — can you see the Pythagorean triple before computing, by drawing the displacement vector?** The displacement $(4, -3)$ is a reordered 3-4-5.

5. Find the midpoint of the segment joining $(4, 8)$ and $(10, 2)$.

Answer: $(7, 5)$

Method: Midpoint averages each coordinate: $(\frac{4+10}{2}, \frac{8+2}{2}) = (7, 5)$.

Investigate further: Check both halves are equal distances: $(7-4, 8-5) = (3, 3)$ and $(10-7, 5-2) = (3, 3)$ ✓. **The midpoint of two points is the average of the two displacement vectors. What is the midpoint of $(4, 8)$ and $(-6, -2)$?** Negative coordinates behave the same way.

6. Find the perpendicular distance from the origin to the line $3x + 4y - 10 = 0$.

Answer: 2

Method: Use $\frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$ at $(0, 0)$: $\frac{|-10|}{\sqrt{9+16}} = \frac{10}{5} = 2$.

Investigate further: The denominator 5 is the length of the normal vector $(3, 4)$. **The point of the line closest to the origin is a multiple $t(3, 4)$ of the normal — find it and confirm its distance is indeed 2.**

7. A line has x -intercept 4 and y -intercept 6. Find its gradient.

Answer: $-\frac{3}{2}$

Method: The intercepts are the points $(4, 0)$ and $(0, 6)$: $m = \frac{6-0}{0-4} = -\frac{3}{2}$.

Investigate further: Reading right-to-left: this is the negative of $\frac{y\text{-intercept}}{x\text{-intercept}}$. **Write the line in intercept form $\frac{x}{4} + \frac{y}{6} = 1$ and rearrange to confirm $m = -\frac{y\text{-intercept}}{x\text{-intercept}} = -\frac{6}{4} = -\frac{3}{2}$ — the intercept form makes the gradient transparent.**

8. Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$.

Answer: 6

Method: Right-angled at the origin: area = $\frac{1}{2} \cdot 4 \cdot 3 = 6$.

Investigate further: Equivalently the shoelace/midpoint-vector formula $\frac{1}{2}|(4)(3) - (0)(0)| = 6$. **Rotate this triangle by 90° about the origin: does the area change, and do you get the same shoelace number?** Reflections and rotations preserve area.

9. The points $(1, 2)$, $(2, 4)$ and $(k, 10)$ are collinear. Find k .

Answer: 5

Method: Rise per unit run is 2 (from $(1, 2)$ to $(2, 4)$); continuing to $y = 10$ climbs 6 more, so $k = 2 + \frac{6}{2} = 5$.

Investigate further: Formal check: $\frac{4-2}{2-1} = \frac{10-4}{k-2} \Rightarrow 2 = \frac{6}{k-2} \Rightarrow k = 5$. **What happens to k if the points are required to be collinear but the order is $(2, 4)$, $(1, 2)$, $(k, 10)$?** Collinearity is order-independent — all three gradients must agree.

10. Using the shoelace formula, find the area of the triangle with vertices $(0, 0)$, $(3, 4)$ and $(6, 0)$.

Answer: 12

Method: Shoelace: $\frac{1}{2}|0 \cdot 4 + 3 \cdot 0 + 6 \cdot 0 - (0 \cdot 3 + 4 \cdot 6 + 0 \cdot 0)| = \frac{1}{2}|24| = 12$.

Investigate further: Bonus route: base = 6, height = 4, area = $\frac{1}{2} \cdot 6 \cdot 4 = 12$ ✓. **For a triangle with one vertex at the origin, shoelace collapses to $\frac{1}{2}|x_1y_2 - x_2y_1|$ with the other two points — prove that this equals $\frac{1}{2}|\det|$ of the two vectors.** That determinant is the signed area.

Section B Manipulation Drills

1. The line $2x - 3y + 6 = 0$ is perpendicular to which of the following?

- A) $y = -\frac{3}{2}x + 1$
 B) $y = \frac{3}{2}x - 2$
 C) $y = \frac{2}{3}x + 5$
 D) $y = -\frac{2}{3}x + 3$

Answer: A) $y = -\frac{3}{2}x + 1$

Method: Given line has gradient $\frac{2}{3}$ (rule $-a/b$), so perpendicular gradient is $-\frac{3}{2}$. Only option A has gradient $-\frac{3}{2}$.

Investigate further: Each decoy makes one half of the flip-and-negate move: option B inverts without negating, option D negates without inverting, and option C is parallel to the given line. **Which option would be perpendicular to $3x + 2y = 1$?**

2. What is the perpendicular distance from $(2, 3)$ to the line $5x + 12y - 7 = 0$?

- A) 1
 B) 2
 C) 3
 D) 4

Answer: C) 3

Method: $\frac{|5(2)+12(3)-7|}{\sqrt{25+144}} = \frac{|10+36-7|}{13} = \frac{39}{13} = 3$.

Investigate further: The numerator is exactly how far the point sits from the line measured along a normal whose length is 13. **Two points are at perpendicular distance 3 from $5x + 12y - 7 = 0$ — what do they have in common?** They lie on $5x + 12y - 7 = \pm 39$, the two parallel satellite lines.

3. Which point lies on the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$?

- A) $(2, 5)$
 B) $(3, 3)$
 C) $(4, 4)$
 D) $(5, 3)$

Answer: A) $(2, 5)$

Method: Midpoint $(3, 4)$; segment gradient 1, so bisector gradient -1 : $y - 4 = -(x - 3)$, i.e. $x + y = 7$.
Check options: $2 + 5 = 7$ ✓.

Investigate further: Only A satisfies $x + y = 7$. **A point is on the perpendicular bisector iff it is equidistant from the two endpoints — verify $(2, 5)$ is distance $\sqrt{16}$ from both $(1, 2)$ and $(5, 6)$.** That's the defining property, worth internalising.

4. Find the area of the quadrilateral with vertices $(0, 0)$, $(4, 0)$, $(5, 3)$, $(1, 3)$, in that order.

- A) 10
B) 12
C) 14
D) 16

Answer: B) 12

Method: Shoelace in order: $\frac{1}{2}|0 \cdot 0 + 4 \cdot 3 + 5 \cdot 3 + 1 \cdot 0 - (0 \cdot 4 + 0 \cdot 5 + 3 \cdot 1 + 3 \cdot 0)| = \frac{1}{2}|0 + 12 + 15 + 0 - (0 + 0 + 3 + 0)| = \frac{1}{2}|24| = 12$.

Investigate further: The quadrilateral is a parallelogram (bases $\parallel$ via position vectors $(4, 0)$ and $(1, 3)$), so area = $|\det| = |4(3) - 0(1)| = 12$ ✓. **Reorder the vertices as $(0, 0)$, $(5, 3)$, $(4, 0)$, $(1, 3)$: the same region, but the shoelace number warns you when points are listed in the wrong order.** Keep vertices in cyclic order.

5. The line through $(1, 2)$ and $(4, 5)$ crosses the y -axis at which value?

- A) 0
B) 1
C) 2
D) 3

Answer: B) 1

Method: Gradient $\frac{5-2}{4-1} = 1$; $y = x + c$ through $(1, 2)$ gives $c = 1$.

Investigate further: Extension: at $x = 0$, $y = 1$. **Through the point-slope form, $y - 2 = 1(x - 1)$. If the line were instead the one through $(1, 2)$ and $(4, 8)$, the y -intercept changes — resolve c again.** Always plug the known point in, never guess c from the picture.

6. Find the perpendicular distance from $(0, 1)$ to the line $6x - 8y + 2 = 0$.

Answer: $\frac{3}{5}$

Method: $\frac{|6(0) - 8(1) + 2|}{\sqrt{36 + 64}} = \frac{|-6|}{10} = \frac{3}{5}$.

Investigate further: Normal length $\sqrt{36 + 64} = 10$; numerator 6. **The line $6x - 8y + 2 = 0$ sits distance $\frac{1}{5}$ from the origin (check $\frac{|2|}{10}$). Since the y -axis point $(0, 1)$ is 1 unit up from the origin, the net distance $|1 - \frac{1}{5}| = \frac{4}{5}$ differs from $\frac{3}{5}$ — why?** Because the normal $(6, -8)$ is not vertical: only the projection along it counts.

7. Which of the following is NOT parallel to $y = 3x + 1$?

- A) $y = 3x + 5$
 B) $6x - 2y = 4$
 C) $3x - y = 2$
 D) $y = 2x + 1$

Answer: D) $y = 2x + 1$

Method: Parallel $\Leftrightarrow$ same gradient 3. A: $3\checkmark$; B: $6x - 2y = 4 \Rightarrow y = 3x - 2\checkmark$; C: $3x - y = 2 \Rightarrow y = 3x - 2\checkmark$; D: $y = 2x + 1$ has gradient 2.

Investigate further: Options B and C are disguises (divide by the y -coefficient first). **Show two distinct parallel lines never intersect, while any two non-parallel lines always do.** Parallelism is exactly “same slope, different intercept”.

8. The line $y = x + 1$ meets the line through $(0, 3)$ perpendicular to it at the point P . Find the coordinates of P .

Answer: ”(1, 2)”

Method: Through $(0, 3)$ with gradient -1 : $y = -x + 3$. Intersect: $x + 1 = -x + 3 \Rightarrow x = 1$, then $y = 2$.

Investigate further: The intersection of two perpendicular lines is the right-angle vertex of a right triangle. Verify $P = (1, 2)$ lies on both lines: $y = x + 1$ gives $2 = 1 + 1 \checkmark$ and $y = -x + 3$ gives $2 = -1 + 3 \checkmark$. **Through $(0, 3)$ draw the perpendicular and reflect the point where it meets $y = x + 1$ across that meeting point — what do you get, and how does it connect to the midpoint of $(0, 3)$ and $(2, 1)$?** Perpendicular lines meet exactly once; the reflect-and-extend trick answers the follow-up.

9. Which point is closest to the origin?

- A) $(1, 3)$
 B) $(2, 2)$
 C) $(4, 0)$
 D) $(0, 5)$

Answer: B) $(2, 2)$

Method: Distances from the origin: $(1, 3)$ gives $\sqrt{10}$, $(2, 2)$ gives $\sqrt{8}$, $(4, 0)$ gives 4, $(0, 5)$ gives 5; the smallest is $\sqrt{8}$.

Investigate further: Comparing squared distances avoids surds: 10, 8, 16, 25. **Among points with x, y nonnegative integers and $x + y = 4$, which is closest to the origin?** Both squares and sum compare against $(2, 2)$.

10. Using the shoelace formula, find the area of the quadrilateral with vertices $(1, 0)$, $(5, 2)$, $(4, 6)$, $(0, 3)$, in that order.

Answer: $\frac{33}{2}$

Method: Cross terms $x_i y_{i+1} - x_{i+1} y_i$ in order: $1 \cdot 2 - 5 \cdot 0 = 2$, $5 \cdot 6 - 4 \cdot 2 = 22$, $4 \cdot 3 - 0 \cdot 6 = 12$, $0 \cdot 0 - 1 \cdot 3 = -3$. Their sum is 33, so the area is $\frac{33}{2}$.

Investigate further: List the same four vertices in the order $(1, 0), (4, 6), (5, 2), (0, 3)$. The shoelace sum now describes a self-crossing path: what does it give, and why is it not an area?

Section C Substitution & Structure

(≤ 90 s each)

Competition-hard: hidden structure, non-routine use of the same tools. No question is a direct formula plug.

1. The line $l: y = 6 - 2x$ meets the x -axis at A . A line m , perpendicular to l , meets the negative x -axis at $(k, 0)$. The lines l , m and the x -axis enclose a triangle of area 20. What is k ?

- A) -7
 B) -10
 C) $3 - 5\sqrt{2}$
 D) $3 - 2\sqrt{5}$

Answer: A) -7

Method: $A = (3, 0)$, and m has gradient $\frac{1}{2}$: $y = \frac{1}{2}(x - k)$. It meets l where $x = \frac{12+k}{5}$, at height $\frac{2}{5}(3 - k)$. The base is $3 - k$, so the area is $\frac{1}{5}(3 - k)^2 = 20$, giving $3 - k = 10$ and $k = -7$. Faster: the triangle is right-angled with legs in the ratio $1 : 2$, so its area is always $\frac{1}{5}$ of the base squared. Option B measures the base from the origin; option C drops the $\frac{1}{2}$; option D drops the $\frac{1}{5}$.

Investigate further: Without the word *negative* a second value of k works. Find it, and explain where that triangle lies.

2. The perpendicular bisector of the segment joining $A(2, -6)$ and $B(b, 4)$, where $b > 0$, passes through the origin. What is b ?

- A) $2\sqrt{10}$
 B) $2\sqrt{14}$
 C) 12
 D) $2\sqrt{6}$

Answer: D) $2\sqrt{6}$

Method: A point lies on the perpendicular bisector of AB exactly when it is equidistant from A and B . So $OA^2 = OB^2$: $4 + 36 = b^2 + 16$, $b^2 = 24$ and $b = 2\sqrt{6}$. No gradients needed. Option A is OA itself; option B adds the 16 instead of subtracting it; option C makes $\angle AOB$ a right angle instead.

Investigate further: For which b does the perpendicular bisector pass through $(1, 0)$? Explain, using a circle and the line $y = 4$, why any given point gives at most two values of b .

3. The point $A(2, 9)$ is reflected in the line $x + 2y = 5$ to give the point A' . How far is A' from the origin?

- A) $\sqrt{85}$
 B) $\sqrt{10}$
 C) 5
 D) $\sqrt{185}$

Answer: C) 5

Method: Move along the normal $(1, 2)$. At A , $x + 2y - 5 = 15$, and each step of $(1, 2)$ changes it by $1^2 + 2^2 = 5$. Reaching the line takes 3 steps back (the foot is $(-1, 3)$); reflecting takes 6: $A' = (2, 9) - 6(1, 2) = (-4, -3)$, so $|OA'| = 5$. Option B stops at the foot; option A forgets to move at all; option D uses $+5$ instead of -5 for the constant.

Investigate further: Find the point of the line $x + 2y = 5$ closest to the origin. Use it to explain why $|OA'|$ and $|OA|$ differ, even though a reflection preserves distances to every point of the mirror.

4. The lines $3x - 4y + 2 = 0$ and $4x - 3y - 5 = 0$ cross, dividing the plane into four angular regions. Which line bisects the angle whose region contains the point $(1, 1)$?

- A) $x - y = 7$
 B) $x + y = 7$
 C) $7x + 7y = 3$
 D) $7x - 7y = 3$

Answer: D) $7x - 7y = 3$

Method: Both normals have length 5, so the bisectors are $(3x - 4y + 2) = \pm(4x - 3y - 5)$. At $(1, 1)$ the two left-hand sides are 1 and -4 : opposite signs. Points in that region satisfy $3x - 4y + 2 = -(4x - 3y - 5)$, i.e. $7x - 7y = 3$. Option B is the other bisector, the one for the regions where the signs agree.

Investigate further: Which of the two bisectors cuts the *acute* angle between the lines? Decide using the sign of the dot product of the normals $(3, -4)$ and $(4, -3)$.

5. The points $A(1, 4)$, $B(4, 7)$ and $C(10, k)$ are collinear. The line through C perpendicular to AB meets the y -axis at D . What is the y -coordinate of D ?

- A) 23
 B) 24
 C) 25
 D) 26

Answer: A) 23

Method: AB has gradient 1, so $k = 4 + 9 = 13$. The perpendicular through $C(10, 13)$ has gradient -1 : $y = 23 - x$, which meets the y -axis at 23.

Investigate further: If B were $(4, 10)$ instead, AB has gradient 2. Show that $k = 22$ and D becomes $(0, 27)$, then find D in general when AB has gradient m .

6. Triangle ABC has vertices $A(0, 0)$, $B(9, 3)$ and $C(3, 9)$. The point P lies on AB with $AP : PB = 1 : 2$, and Q lies on AC with $AQ : QC = 2 : 1$. What is the area of the quadrilateral $PBCQ$?

- A) 24
 B) 28
 C) 27

D) 32

Answer: B) 28

Method: $[ABC] = \frac{1}{2}|9 \cdot 9 - 3 \cdot 3| = 36$. Triangle APQ shares angle A , so $[APQ] = \frac{1}{3} \cdot \frac{2}{3} \cdot 36 = 8$, and $[PBCQ] = 36 - 8 = 28$. Option D uses $\frac{1}{3}$ for both ratios; option C treats P and Q as midpoints; option A keeps only one ratio.

Investigate further: For which ratio $AQ : QC$ (keeping $AP : PB = 1 : 2$) does $PBCQ$ have exactly three times the area of APQ ?

7. The lines $y = 2x + 1$, $y = 7 - x$ and $y = \frac{1}{2}x - 2$ enclose a triangle. What is its area?

A) 12

B) 18

C) 24

D) 48

Answer: C) 24

Method: Pairwise intersections: $(2, 5)$, $(6, 1)$ and $(-2, -3)$. Shoelace: $\frac{1}{2}|(2-30)+(-18+2)+(-10+6)| = \frac{1}{2} \cdot 48 = 24$. Option D forgets the $\frac{1}{2}$; option A halves twice.

Investigate further: The vertices $(2, 5)$ and $(-2, -3)$ lie on $y = 2x + 1$. Use that side as the base and the perpendicular distance from $(6, 1)$ as the height, and check that you get 24 again.

8. The quadrilateral Q has vertices $(0, 0)$, $(4, 0)$, $(5, 3)$ and $(1, 3)$ in that order (a parallelogram). A second quadrilateral Q' is obtained by reflecting Q in the line $y = x$. What is the area of the overlap $Q \cap Q'$?

A) 6

B) 8

C) 9

D) 12

Answer: A) 6

Method: Q is $0 \leq y \leq 3$, $y \leq x \leq y + 4$. Reflecting swaps x and y , so Q' is $0 \leq x \leq 3$, $x \leq y \leq x + 4$. Both hold on the quadrilateral $(0, 0)$, $(3, 1)$, $(3, 3)$, $(1, 3)$: the square $[0, 3]^2$ (area 9) minus the triangles $(0, 0)$, $(3, 0)$, $(3, 1)$ and $(0, 0)$, $(0, 3)$, $(1, 3)$ (each $\frac{3}{2}$), so $9 - 3 = 6$. Option C forgets to remove the triangles; option D is the area of Q itself.

Investigate further: Slide Q one unit to the right, to vertices $(1, 0)$, $(5, 0)$, $(6, 3)$, $(2, 3)$. Reflect it in $y = x$ again and find the new overlap.

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Hardest 5: deep insight, hidden structure, or a construction you must discover. No diagram gives away the trick.

1. The points $A(1, 3)$ and $B(7, 5)$ are fixed, and P moves along the line $y = x$. What is the least possible value of $PA + PB$?

- A) $4\sqrt{2}$
- B) $2\sqrt{34}$
- C) 10
- D) $2\sqrt{10}$

Answer: D) $2\sqrt{10}$

Method: Check sides first: A has $y > x$ and B has $y < x$, so they are on *opposite* sides of $y = x$. The segment AB crosses the line at $(4, 4)$, and $PA + PB \geq AB = \sqrt{36 + 4} = 2\sqrt{10}$, with equality there. Option A reflects A anyway (the reflection trick is only needed when both points are on the same side); option C reflects in the x -axis; option B reflects through the origin.

Investigate further: Move B to $(5, 7)$, on the same side as A . Now the reflection is needed: find the least value of $PA + PB$ and the point P that achieves it.

2. The square $MNOP$ has side 10. Points R, S, T, U lie on MN, NO, OP, PM respectively, and $RSTU$ is a rectangle whose sides make 45° with the sides of the square. The area of $RSTU$ is 20. What is the length of a diagonal of $RSTU$?

- A) $4\sqrt{15}$
- B) $4\sqrt{10}$
- C) 10
- D) $4\sqrt{5}$

Answer: B) $4\sqrt{10}$

Method: Put $M = (0, 0)$ and $R = (x, 0)$. The 45° sides force $S = (10, 10 - x)$ and $U = (0, x)$, so $RS = \sqrt{2}(10 - x)$ and $RU = \sqrt{2}x$. The area gives $2x(10 - x) = 20$, i.e. $x(10 - x) = 10$; there is no need to find x . The diagonal satisfies $d^2 = RS^2 + RU^2 = 2[(x + (10 - x))^2 - 2x(10 - x)] = 2(100 - 20) = 160$, so $d = 4\sqrt{10}$. Option C assumes $RSTU$ is a square; option D forgets the $\sqrt{2}$ on each side; option A adds $2x(10 - x)$ instead of subtracting it.

Investigate further: Show that the perimeter of $RSTU$ is the same for every position of R . What is the largest possible area, and which rectangle has it?

3. What is the area of the region of points (x, y) satisfying both $|x| + 2|y| \leq 6$ and $y \geq x - 2$?

- A) 10
- B) 18
- C) 26
- D) 36

Answer: C) 26

Method: $|x| + 2|y| \leq 6$ is the rhombus with vertices $(\pm 6, 0)$, $(0, \pm 3)$, of area $\frac{1}{2} \cdot 12 \cdot 6 = 36$. The line $y = x - 2$ cuts its edges at $(-\frac{2}{3}, -\frac{8}{3})$ and $(\frac{10}{3}, \frac{4}{3})$. The part *below* the line is the quadrilateral $(-\frac{2}{3}, -\frac{8}{3}), (0, -3), (6, 0), (\frac{10}{3}, \frac{4}{3})$, with shoelace area 10. So the required area is $36 - 10 = 26$. Option A is the piece cut off; option B assumes the line halves the rhombus.

Investigate further: For which c does the line $y = x + c$ split the rhombus into two equal areas? Explain why symmetry forces the answer.

4. How many straight lines through the point $(2, 3)$ form, together with the two coordinate axes, a triangle of area 12?

- A) 1
B) 2
C) 4
D) 3

Answer: D) 3

Method: A line of gradient m through $(2, 3)$ has intercepts $2 - \frac{3}{m}$ and $3 - 2m$, so the area is $\frac{(2m - 3)^2}{2|m|}$. Setting it to 12: for $m > 0$, $4m^2 - 36m + 9 = 0$ has two positive roots; for $m < 0$, $4m^2 + 12m + 9 = (2m + 3)^2 = 0$ gives $m = -\frac{3}{2}$ once. That is 3 lines. Option B misses the negative branch; option C counts the double root twice.

Investigate further: The value 12 is exactly the *smallest* area of a first-quadrant triangle through $(2, 3)$, which is why the negative branch has a double root. Prove that the minimum occurs when $(2, 3)$ is the midpoint of the hypotenuse.

5. For each t with $0 < t < 4$, the line L_t passes through $(t, 0)$ and $(0, 4 - t)$. What is the greatest possible distance from the origin to L_t ?

- A) 1
B) 2
C) $\sqrt{2}$
D) $2\sqrt{2}$

Answer: C) $\sqrt{2}$

Method: With $a = t$, $b = 4 - t$ the distance is $d = \frac{ab}{\sqrt{a^2 + b^2}}$, and $a^2 + b^2 = 16 - 2ab$. So $d^2 = \frac{p^2}{16 - 2p}$ with $p = ab$, which increases with p . The product is largest ($p = 4$) at $a = b = 2$, giving $d^2 = 2$ and $d = \sqrt{2}$. Option B reports d^2 ; option D is the distance from the origin to the midpoint $(2, 2)$ of the segment, not to the line.

Investigate further: Every L_t is tangent to one fixed curve, the envelope of the family. Find it, and check that the foot of the perpendicular from the origin to L_2 lies on it.

Top 5 Patterns — Worksheet #1

- Implicit line $\rightarrow$ gradient** $-a/b$. Read it off $ax + by + c = 0$ without rearranging; the constant never touches the gradient. (see A1, A7, B7)
- Perpendicular bisector = equidistance.** $PA^2 = PB^2$ is one linear equation and needs no gradient at all. (see B3, C2, C5)
- Move along the normal.** Distance to a line, the foot of a perpendicular and a reflection all come from stepping along (a, b) from the point. (see A6, B2, C3, C4)

4. **Shoelace in cyclic order.** Half the absolute cross-term sum; vertices must go round the shape in order. *(see A10, B4, B10, C7, D3)*
5. **Area by ratio before area by coordinates.** Intercepts, shared angles and ratios give most areas in one line. *(see C1, C6, D2, D4)*

Common Traps — Worksheet #1

1. **Half a perpendicular flip.** $m = \frac{2}{3}$ pairs with $-\frac{3}{2}$: invert *and* negate. *(see A3, B1)*
2. **Stopping at the foot.** A reflection travels twice as far along the normal as the foot of the perpendicular. *(see C3)*
3. **Picking the wrong bisector.** Two lines have two angle bisectors; the signs of the line expressions at a test point decide which one you want. *(see C4)*
4. **Reflecting when you don't need to.** If the two points are on opposite sides of the line, the straight segment already crosses it. *(see D1)*
5. **Losing a branch.** Absolute values and sign conditions hide extra solutions: check every sign case before counting, and use the stated condition to pick one. *(see C1, D4)*

“Shoelace before altitude: area is a one-line cross product, not a construction.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths